



ASSIGNMENT

Marks		Course No.	EEE 4109
Assigned	Obtained	Course Title	Microprocessors and Embedded Systems
10		Course Instructor(s)	Asif Zaman Rizve, Lecturer, Dept. of EEE, RUET Md. Ariful Islam, Lecturer, Dept. of EEE, RUET
		Student Name	Mir Masrur
Due Date	28-10-2025	Roll No.	2001019
Submission Date	05-11-2025	Section	A

Problem Statement:

Design a PLC ladder diagram and hardware diagram for a Multi-Zone Smart Irrigation System. The system monitors three garden zones using soil moisture sensors (S1, S2, S3) and includes a rain detector (R) and a manual override switch (M). The outputs consist of solenoid valves (V1, V2, V3) and a water pump (P). The system operates such that when any zone's moisture sensor detects dryness (logic 0), the corresponding valve opens and the pump starts. If rainfall is detected ($R = 1$), irrigation is immediately halted. Once all zones have sufficient moisture (all $S = 1$), the pump automatically turns off. Additionally, the manual override switch M enables the user to start the pump and all valves for 10 minutes, regardless of the sensor readings.

The solution of the assignment must contain:

1. **Problem Statement:** Provide a concise description of the problem to be addressed.
2. **Hardware Diagram:** Provide a clearly labeled and detailed hardware diagram of the system. Generate the diagram digitally and include a high-resolution image of it.
3. **Ladder Diagram:** Provide a clearly labeled and detailed ladder diagram for the operational requirements of the system. Simulate the ladder diagram in Logosoft Comfort or CX programmer software and attach a high-resolution image of it.
4. **Results and Discussion:** Attach the simulation results of the simulated ladder diagram. Also, discuss how the results satisfy the problem requirements.
5. **Addressing Course Outcomes (COs) and Program Outcomes (POs):** Explain how your solution addresses CO3 and PO(c).
6. **Addressing Complex Engineering Problems:** Justify how your solution satisfies P1 (Depth of knowledge), P2 (Range of conflicting requirements), and P3 (Depth of analysis required).

Note: Any plagiarized submission will result in a zero mark.



Attribute	Justification
CO3: Apply the knowledges for solving different real-life complex engineering problems.	The provided PLC problem requires to apply the knowledge of automation, sensors, actuators, and control logic to design and implement a working system that operates under real-world conditions.
PO(c): Design/Development of Solution	The provided PLC problem enables students to design and develop practical automation solutions using sensors, actuators, and control logic. It helps them apply theoretical knowledge to create efficient and reliable systems that address real-world engineering problems.

Addressing Complex Engineering Problems (P's):

Attribute	Justification
P1: Depth of Knowledge required	The provided PLC problem requires in-depth knowledge of PLC operation, sensors, and control logic to design an automated system.
P2: Range of conflicting requirements	The task involves balancing multiple conditions such as timing, safety, and user control.
P3: Depth of analysis required	Students must analyze system behavior, simulate operations, and troubleshoot logic to ensure correct sequence and reliable performance.

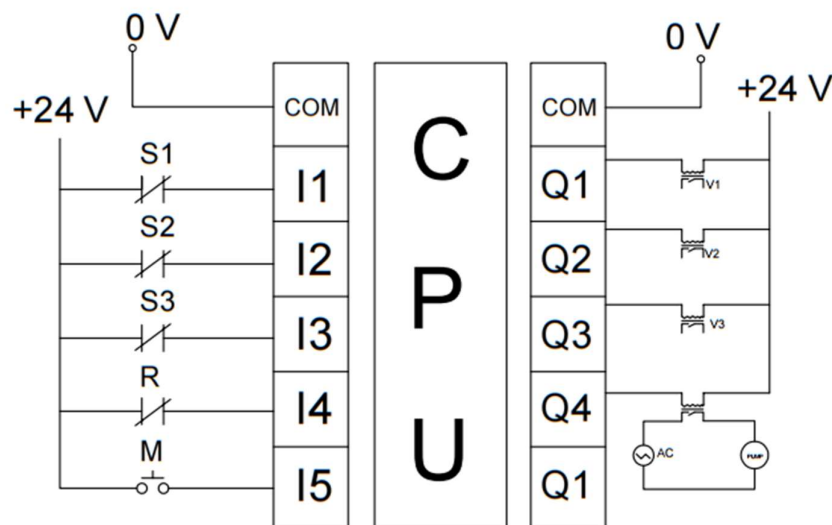
Rubrics:

Source	Marks	Corr. CO	Distribution of Marks				Obtained Marks
Hardware Diagram	3	CO3	3: Diagram with proper connections and labelling	2: Diagram with proper connection, but improper labelling	1: Diagram with slightly incorrect connection	0: Diagram with mostly incorrect connection	
Ladder Diagram	3	CO3	3: Diagram with proper connections and labelling	2: Diagram with proper connection, but improper labelling	1: Diagram with slightly incorrect connection	0: Diagram with mostly incorrect connection	
Results and Discussion	2	CO3	2: Results are correct and properly presented with adequate explanation.	1: Results are slightly incorrect or not properly presented with adequate explanation	0: Results are mostly incorrect and not properly presented with adequate explanation		
Addressing CO and PO	1	CO3	1: Clearly justifies how CO3 and PO(c) are addressed.	0: Unclear justification or justification not addressed.			
Addressing P's	1	CO3	1: Clearly justifies how P1, P2 and P3 are addressed.	0: Unclear justification or justification not addressed			
Total Obtained Marks (out of 10)							

Problem Statement

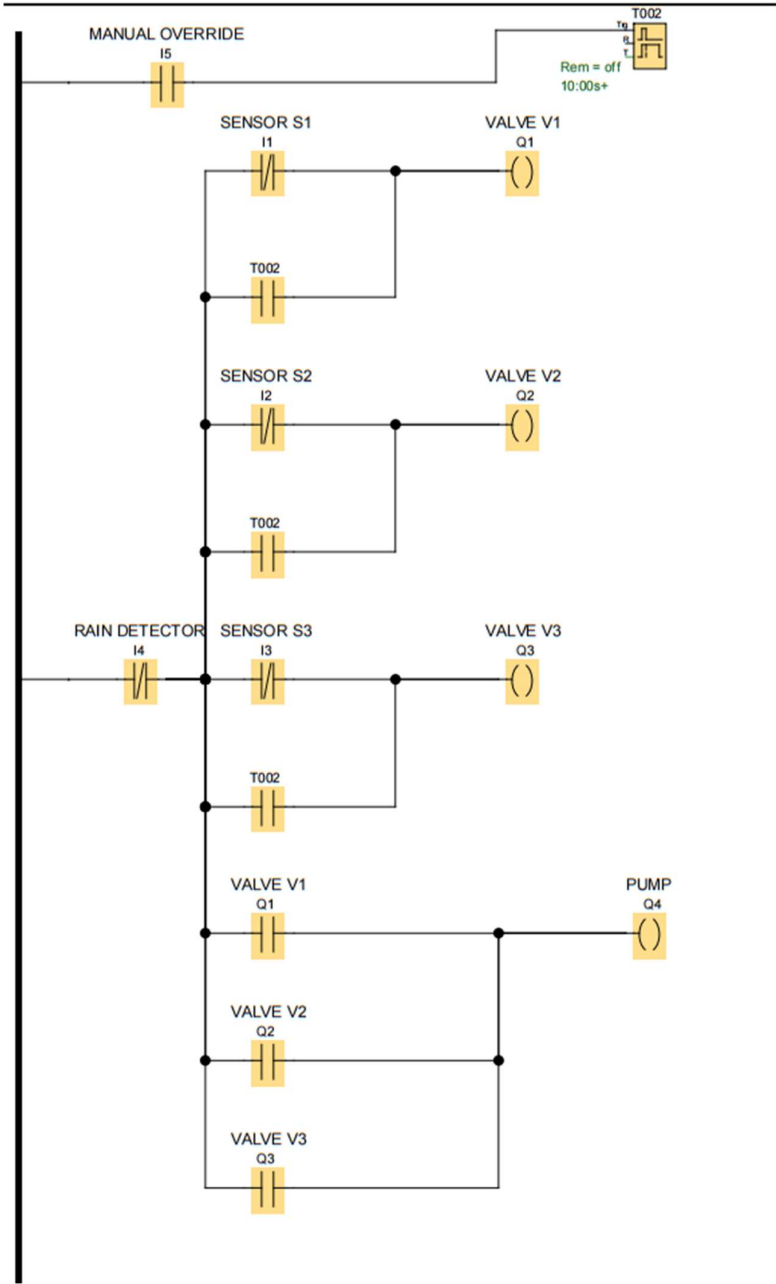
Design a PLC ladder diagram and hardware diagram for a Multi-Zone Smart Irrigation System. The system monitors three garden zones using soil moisture sensors (S1, S2, S3) and includes a rain detector (R) and a manual override switch (M). The outputs consist of solenoid valves (V1, V2, V3) and a water pump (P). The system operates such that when any zone's moisture sensor detects dryness (logic 0), the corresponding valve opens and the pump starts. If rainfall is detected ($R = 1$), irrigation is immediately halted. Once all zones have sufficient moisture (all $S = 1$), the pump automatically turns off. Additionally, the manual override switch M enables the user to start the pump and all valves for 10 minutes, regardless of the sensor readings.

Hardware Diagram



This diagram shows the essential PLC hardware structure, connecting field devices to the CPU. On the left, the input module (I1-I5) receives signals from sensors (S1, S2, S3, R) and a Manual Override switch (M), operating on a +24 V DC source in a sourcing configuration. The central CPU processes these input states according to a stored program. On the right, the output module (Q1-Q4) controls external valves. Outputs Q1, Q2, and Q3 energize +24 V DC control valves (V1, V2, V3), while Q4 (pump) interfaces with an AC-powered circuit to activate a pump, utilizing intermediate control elements for switching and isolation between the low-voltage control side and the higher-power load side.

Ladder Diagram



Result and Discussion

Check Conditions:

- When all the sensor has high input as $S=1$, pump is off automatically:



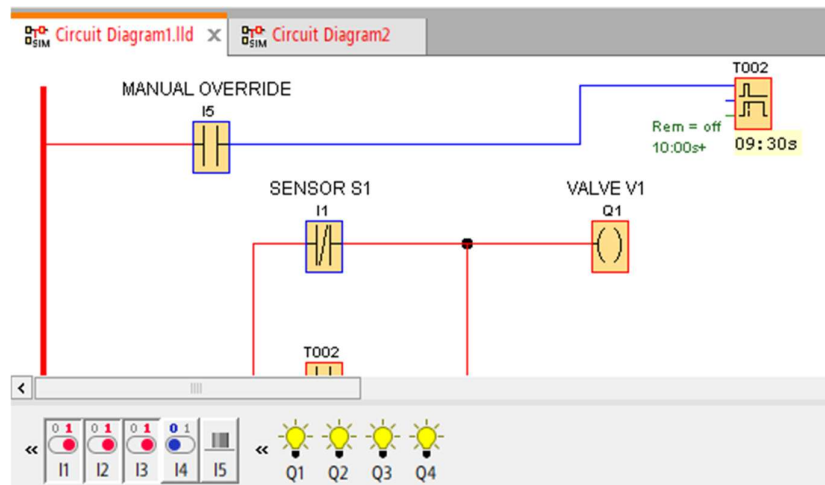
- When any of the sensors detect dryness as $S=0$, corresponding valves and pump get starts:



- If rainfall detected, $R=1$, all valves and pump turns off



- When manual override switch M enables its start the pump and all valves for 10 minutes (10 sec shows in diagram for better understandings), regardless of the sensor readings.



Addressing Course Outcomes (COs) and Program Outcomes (POs):

My solution address

CO3: Apply the knowledges for solving different real-life complex engineering problems &

PO(c): Design/Development of Solution.

CO3:

My solution applies my knowledge of PLC operation and control logic. It takes four different sensor output as input of the PLC and controls corresponding valves and a pump, which is a real life irrigation problem solution. My ladder logic implements complex logic like automatic irrigation based on dryness (S=0), immediate shutdown on rain (R=1), and a 10 Minutes manual override.

PO(c):

I designed and developed both the detailed Hardware Diagram and the complete Ladder Diagram, that satisfy this PO as well.

Addressing Complex Engineering Problems

Attribute	Justification of My Solution
P1: Depth of Knowledge required	My solution required in-depth knowledge of PLC programming, sensor logic (dryness = Logic 0, and valve control to design a automated system.
P2: Range of conflicting requirements	I managed conflicting requirements by balancing automatic operation (sensors), safety (rain detection R=1 immediately halts irrigation and user control (10-minute manual override M.
P3: Depth of analysis required	I had to analyze the system behavior to ensure the correct sequence, such as turning the pump off only when all zones have sufficient moisture (all S=1) and enables the pump and all valves for 10 minutes, regardless of the sensor readings.